

CSE563 — Assignment 1

Unit I: Basic Mathematics in Computer Science

Maps to CO1 (recall the basic concepts of mathematics in computer science and engineering).
Marks: 50. Attempt all questions. Show every step of working; an unsupported final answer earns no marks.

Part A — Short Answer and Definitions (10 marks, 2 each)

Q1. [2 marks] State the division algorithm. Using it, find q and r when $a = -47$ and $d = 9$, and explain why the C expression `-47 % 9` does not return your value of r .

Q2. [2 marks] Let $A = \{1, 2, 3\}$. Write $\mathcal{P}(A)$ in full and state $|\mathcal{P}(A)|$. If $|B| = 6$, how many proper subsets does B have?

Q3. [2 marks] Write the contrapositive, converse and inverse of: “If a graph is a tree, then it is connected.” State which of the three is logically equivalent to the original.

Q4. [2 marks] Define an equivalence relation. Show that congruence modulo 4 on $\mathbb{Z}$ is one, and list its equivalence classes.

Q5. [2 marks] Classify $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^3 - 2$ as injective / surjective / bijective, with justification. Give f^{-1} if it exists.

Part B — Numerical Problems (24 marks)

Q6. [5 marks] Use the Euclidean algorithm to compute $\gcd(1071, 462)$, then use the extended Euclidean algorithm to find integers x, y with $1071x + 462y = \gcd(1071, 462)$. Verify your answer. Finally state $\text{lcm}(1071, 462)$.

Q7. [5 marks] (a) Find $11^{-1} \bmod 26$ using the extended Euclidean algorithm. (b) Compute $5^{20} \bmod 7$ using Fermat's little theorem, and verify by fast modular exponentiation.

Q8. [4 marks] In a survey of 120 students: 65 use Windows, 48 use Linux, 40 use macOS, 22 use Windows and Linux, 18 use Windows and macOS, 15 use Linux and macOS, and 8 use all three. How many use (a) at least one of the three, (b) none of them, (c) exactly one?

Q9. [5 marks] Simplify $F(A, B, C) = \sum m(0, 2, 4, 5, 6)$ using a Karnaugh map. Draw the map,

mark your groups, and write the minimal SOP expression. Verify your answer algebraically for the input $A=1, B=0, C=1$.

Q10. [5 marks] For $f(x, y) = x^2 + 3xy + 2y^2$: (a) find $\partial f / \partial x$ and $\partial f / \partial y$; (b) evaluate the gradient at $(1, 2)$; (c) perform one step of gradient descent from $(1, 2)$ with learning rate $\alpha = 0.1$ and state whether f decreased.

Part C — Proofs and Applied Reasoning (16 marks)

Q11. [5 marks] Prove by mathematical induction that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 1$. State the basis step and the inductive step explicitly.

Q12. [4 marks] Prove by induction that $3 \mid (n^3 - n)$ for every integer $n \geq 0$.

Q13. [4 marks] Prove the set identity $A - (B \cap C) = (A - B) \cup (A - C)$ by the element method (double inclusion). Name the logical law that does the work.

Q14. [3 marks] Draw the Hasse diagram of the poset $(D_{20}, |)$ where $D_{20} = \{1, 2, 4, 5, 10, 20\}$ is the set of positive divisors of 20. Identify the least and greatest elements, and give one pair of incomparable elements together with their greatest lower bound and least upper bound.